

### ■ 1.6.3 Special Topic: Saving Your Definitions

Any function you define in *Mathematica* will last for the whole of your session, unless you explicitly **Clear** it. However, if you want to use the same function again in future sessions, then you explicitly have to save it.

You can save the definitions of functions in exactly the same way as you save definitions of variables.

|                                                                  |                                         |
|------------------------------------------------------------------|-----------------------------------------|
| <code>Save["filename", f<sub>1</sub>, f<sub>2</sub>, ...]</code> | save definitions of functions in a file |
| <code>&lt;&lt;filename</code>                                    | read in a file of definitions           |

Saving and restoring function definitions.

This defines a function `f`.

```
In[1]:= f[x_] := x^2
```

`g` is defined in terms of `f`.

```
In[2]:= g[x_] := 3 f[x] + f[x^2]
```

This saves in the file `gdef.m` the definition of `g`, together with any subsidiary definitions that are needed.

```
In[3]:= Save["gdef.m", g]
```

The file contains the definition of `f` as well as `g`, since `g` depends on `f`.

```
In[4]:= !!gdef.m
g/: g[x_] := 3*f[x] + f[x^2]
f/: f[x_] := x^2
```